

EE302 Lab 11: Buck Converter Final Project

Obtain at checkout:

Multimeter and probes
2x oscilloscope probes
1x BNC to BNC cable
1x small screwdriver (may need to float in lab)

Obtain in lab:

2x CSD18534KCS NMOS FETs
2x Resistor 50 ohm (we should have plenty, but plus or minus 15 ohms or so won't hurt)
1x Resistor 0.5 ohm (high-power)
1x Resistor 5 kohm
1x Resistor 20 kohm
3x Capacitor 10u
4x Capacitor 1u
1x Inductor 100 uH
1x TPS2832 Gate Driver
1x TLV2372 Dual Op Amp
16x Test Point (or more, color coded as desired)
3x Screw Terminals (green)
1x BNC Connector

Motivation:

The final project for EE 302 will be to build a power converter, i.e. an electrical circuit that can provide a desired voltage to a load while receiving a different voltage from a power source. (An example would be a circuit that converts the 110 Vrms sinusoidal voltage of the grid to the 19 Vdc that is compatible to input to your laptop). In particular, we will build a buck converter, which produces a lower output voltage than its input voltage.

We already know some ways to reduce voltages (resistor dividers; certain amplifier configurations), but these are all lossy converters, meaning that they dissipate power. Power converters, by contrast, are theoretically 100% efficient and, in practice, regularly achieve efficiencies of 90+%.

Understanding the buck converter requires you to use what you've learned in Part I (linear systems/circuits), Part II (semiconductor devices, esp MOSFETs), and Part III (frequency domain) of 302.

Prelab:

Navigate to the following link:

<https://www.ti.com/video/series/experimental-power-electronics-curriculum-and-reference.html#>

and watch the videos

- Introduction and Mathematical Tools for Power Electronics > Periodic Steady State (8:19)
- Pulse-Width-Modulated (PWM) Converters > Steady State Analysis (15:04)

Optionally, also this video from Prof. Sam Ben Yaakov:

<https://www.youtube.com/watch?v=HtwilPekfs&t=2s> (until 24:33)

Finally, these two videos on hot-air soldering:

How to Use Hot Air Rework Station (15:00)

<https://www.youtube.com/watch?v=v58m-S35s24> (start at 18:50)

Level Up Your Skills: Hot Air Soldering Made Simple (4:46)

<https://www.youtube.com/watch?v=Xcxp0ofb9OI>

(Total videos duration: 67:42 at default playback speed, including optional)

Questions	Answers
For a buck converter in steady state, what is the output voltage as a function of the input voltage and the duty cycle?	
What is the component law for an inductor?	
Why does the inductor current increase/decrease linearly in the buck converter?	
Why do SMD components “pull into place” as the solder paste melts?	
As a reminder – when should soldering irons be turned off? And what should you do before turning soldering irons off?	

Lab: (you are unlikely to finish in one lab session. be sure to clean up properly each lab day)

An important part of engineering is modularity. That is, complex systems are made up of modules with well-defined input-output characteristics that can be built and tested individually before being assembled into a larger system. We will build the buck converter in stages using modularity as a guiding philosophy to help us ensure that our system works piece by piece, rather than assembling it all and praying that it all works.

1. First, solder the bypass capacitor and the bootstrap capacitor for the gate driver. Also solder the BNC connector and the screw terminal that is meant to provide 10V to the system. Solder the test points for the PWM and LOWDR signals and their corresponding ground test points. Solder both MOSFETs and the resistors in series with their gates (these are 100 ohms in the image in Lab 8, but the system works better with 50 ohms). At some point in this process, go to the TAs for help soldering the gate driver chip itself, which is best done with a hot air gun. Some groups should do this early, and other groups should solder other components first and then do this step afterwards.
2. Connect the 6V power supply output to the screw terminal, carefully paying attention to which connections are positive and negative. Connect the signal generator to the BNC connector.
3. Turn on the function generator and program it to output a square wave with a frequency of 100 kHz. Choose the amplitude and offset to produce a square wave that goes from 0V to 5V. Set the Duty Cycle, i.e. the fraction of the square wave that is high as opposed to low, to 25%.
4. Measure the PWM signal on the PCB to ensure that the signal is indeed making it to the PCB and that it is what you want it to be.
5. Turn on the power supply and set the output to 6V. This will power the gate driver. Check the current from the power supply to ensure that the supply is not current limiting. The voltage should reach 6V and the current should be <10 mA.
6. With the PWM signal running, measure both PWM and LOWDR (the gate drive signal being sent to the lower MOSFET). When PWM is high, LOWDR should be low. Check that this is the case. If so, you can be reasonably confident that the gate driver “module” is working as intended.

If the gate driver gives you trouble, try running the PWM at 50 kHz instead of 100 kHz. If that doesn't solve anything, then you have a different problem. It did help some teams in 2023. The instructions have been fixed such that this frequency reduction should not be necessary, but we will see.

7. Probe the gate pin of the lower MOSFET. You may be able to grab its gate pin, or you can remove the “hat” from the probe and just press it to the pin or a trace. Over a long time scale (several cycles at 100 kHz), you may not be able to tell the difference between LOWDR and Vgs. However, if you zoom in on a rising or falling edge, you may see that LOWDR is snappier than Vgs. This is intentional.
8. Test HIGHDR and LOWDR together to make sure they seem to be switching in a way that will alternately turn Q1 and Q2 on (with Q1 on for duty cycle $D=25\%$). Be careful to note that both MOSFETs are never on simultaneously.

Capture a screenshot of the LOWDR and HIGHDR over several cycles at $V_{in}=0$ and 25% duty cycle, then reduce the power supply voltages to zero and shut it off before continuing.

9. Install the input capacitors (C1, C6, C7 in Lab 8) and the screw terminal meant to provide V_{in+} to the system. Connect the 25V power supply output to the V_{in+} terminal block along with the other connections as before. Set the 6V output to 6V **and the 25V output to zero**. Attach probes to SW and LOWDR. Power on the supply and the output of the function generator. You should see LOWDR switching and SW doing nothing. Now, dial the 25V power supply output to just 3 volts. You should still see LOWDR switching and SW bouncing between 0 and 3V as the high and low MOSFETs switch. Dial the 25V power supply output to 20V and verify that the circuit still works. Reduce both the 25V-rated supply and the 6V-rated supply to zero before disconnecting the circuit.
10. Install the inductor, the output capacitor, and the current sense resistor (0.5 ohm). Repeat the experiment above, measuring SW and V_{out+} (relative to GND). Dial V_{in+} up slowly from 0 as before and ensure that switching looks good. Further, observe V_{out} and make sure that it is about 25% of V_{in} (plus some ripple). You now have a functional buck converter, but you have only tested it at “no load.” You can be reasonably confident that your buck converter works well, but without a load (and therefore no substantial current in any of the components), you still don’t know for sure if it will survive well. We’ll test with a load shortly.

Capture a screenshot of V_{sw} and V_{out} over several cycles with 20V $_{in}$ and 25% duty cycle and “no load.” Then reduce the power supply voltages to zero and shut it off before continuing.

11. Power converters are meant to provide power to do useful work like power a computer or drive a motor. In the lab, we will have to use a “dummy load” consisting of a large resistor. This resistor is only 10 ohm but it is nevertheless very large and attached to a very large heat sink. This is because we plan to dissipate up to 10W in this resistor (again, we are dissipating heat instead of doing useful work because it is easier in the lab).
 - a. If your load is not assembled, you will need a power resistor, a heat sink, a couple of thermal pads, two accompanying screws, and a couple lead wires at least 6” in length. Solder the leads to the ends of the resistor. Place the thermal pads on the bottom of the resistor and then place the entire assembly on the heat sink. Use the screws to make very good contact between the resistor and the heat sink.
 - b. Solder the screw terminal for the load onto the PCB and then connect the load resistor along with all the other connections used previously. Probe SW and V_{out+} and again dial the input voltage up from 0. V_{out+} should still be about 25% of V_{in} , but now substantial current will be flowing in the load and, consequently, the inductor, switches, and input. WARNING – THE LOAD RESISTOR MAY GET HOT. Power off when done.
12. We would like to measure the inductor current in particular. This is the purpose of the current sense resistor, which is in series with the inductor (make sure you understand how it is in series!). The voltage that appears across this resistor is proportional to the current through the inductor, but it is quite a small voltage. The current sense amplifier will help. Solder the dual op amp and the resistors of the inverting amplifier, plus the bypass capacitor. Further solder the OPAMP_OUT and the V_{out-} test points. Probe the SW and V_{out+} points and bring the system back up to full operation (20 V $_{in}$, 25% duty cycle). Now move the probes to V_{out-} and OPAMP_OUT and ensure that OPAMP_OUT is a scaled (and inverted) version of V_{out-} . They should both be approximately triangular with dc offsets. If this is working, then you have a functional buck converter and a means of measuring the main “state variables,” i.e. the output capacitor voltage and the inductor current.

Capture a screenshot of i_L and V_o over several cycles with 20V $_{in}$ and 25% duty cycle. Then reduce the power supply voltages to zero and shut it off before leaving the lab.